

XAI: Open Problems in Mechanistic Interpretability

FAMAF - UNC

First Semester 2026



Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

🍷 Open Problems in Mechanistic Interpretability: A Whirlwind Tour ---



🍷 Motivation ---

Key Question

What should interpretability look like in a post GPT-4 world?

- Large, generative language models are a ****big deal**** - Models will keep scaling — what work done now will matter in the future?
- Emergent capabilities keep arising
- Many mundane problems go away
- A single massive foundation model



🍷 Inputs and Outputs Are Not Enough ---



🍷 Goal: Understand Model Cognition ---

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Is it aligned, or telling us what we want to hear?

🍷 What is a Transformer? ---

- **Input:** Sequences of words - **Output:** Probability distribution over the next word -
- Residual stream:** A sequence of representations - One per input word, per layer - Each layer is an incremental update - Represents the word plus context -
- Attention:** Moves information *between* words - Made of heads, each acts independently -
- MLP:** Processes information *once* it's been moved to a word



🍲 What is Mechanistic Interpretability? ---

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- **Goal:** Reverse engineer neural networks - Like reverse-engineering a compiled binary to source code
- **Hypothesis:** Models learn human-comprehensible algorithms and can be understood, if we learn how to make it legible - Understanding **features** — the variables inside the model - Understanding **circuits** — the algorithms learned to compute features - **Key property:** Distinguishes between cognition with identical output - A deep knowledge of circuits is crucial to understand, predict and **align** model behaviour



🍷 A Growing Area of Research ---



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🍷 Features = Variables: What Does the Model Know? ---

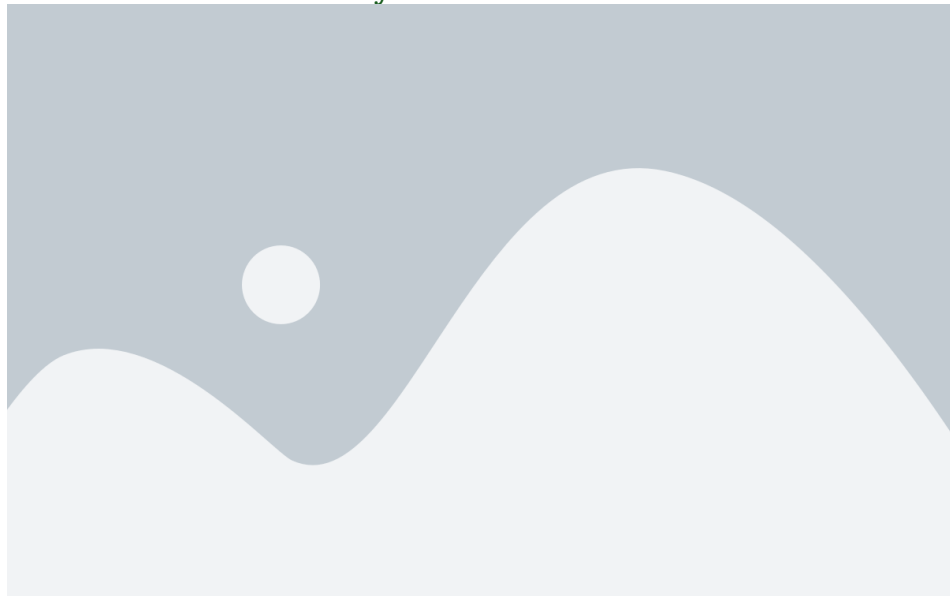


🍷 Features: Linear Representation of Quantities ---

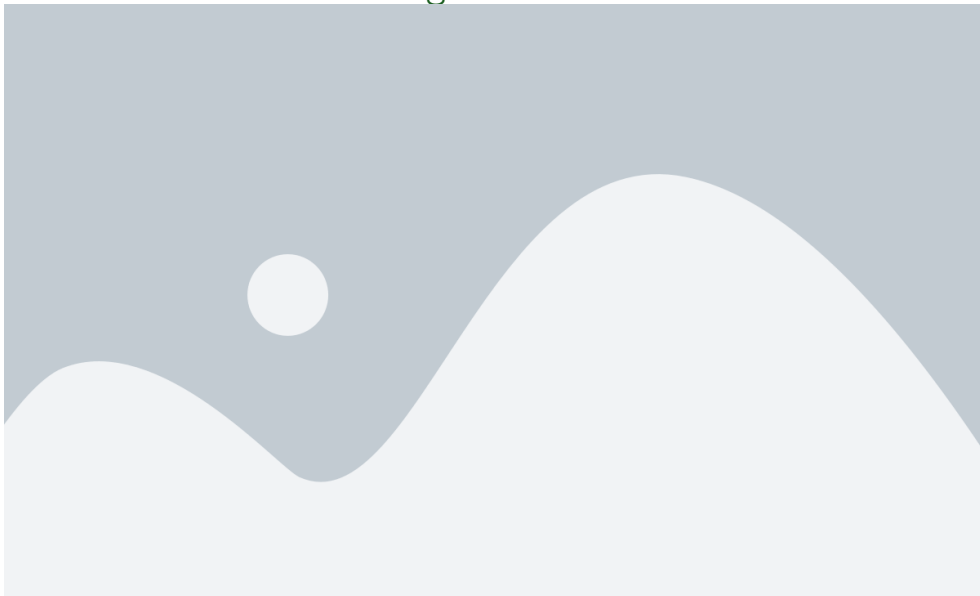


🍷 Circuits = Functions: How Does the Model Think? ---

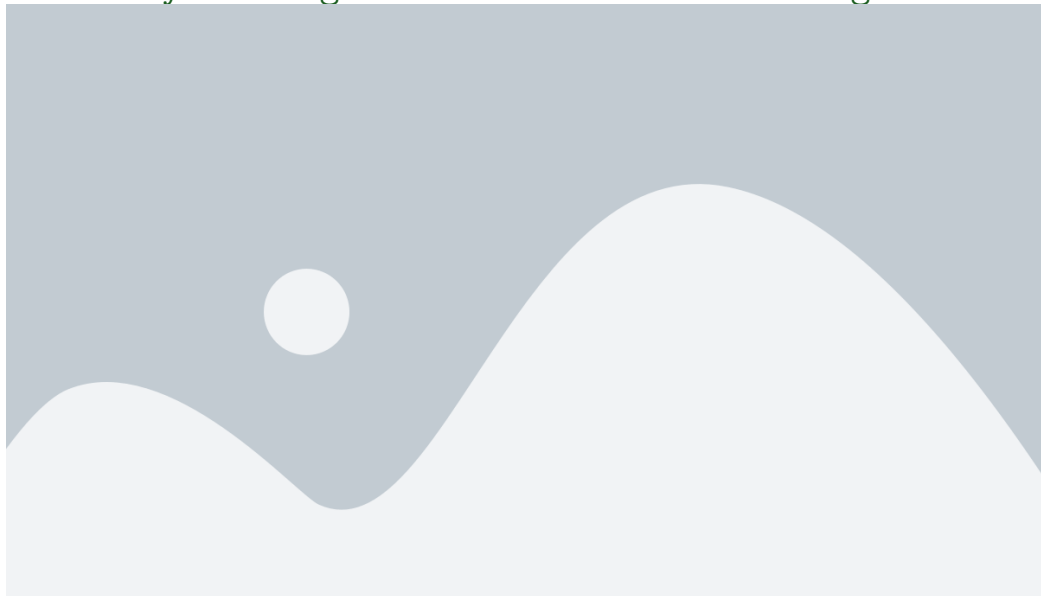
🍷 Induction Heads: A Key Circuit ---



🍲 Mechanistic Understanding of Induction Heads ---



🍷 Case Study: Emergence of In-Context Learning ---



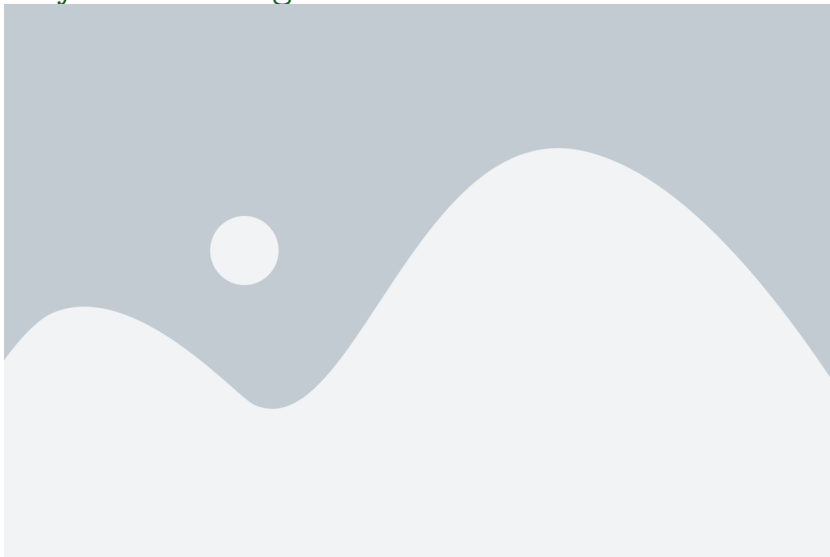
The Mindset of Mechanistic Interpretability ---

- **Alien neuroscience:** Models *are* interpretable, but not in our language — if we learn to think like them, mysteries dissolve
- **Skepticism:** It's extremely easy to trick yourself in interpretability
 - ▶ **Zoom In:** Rigour and depth over breadth and scalability
- **Ambition:** It *is* possible to achieve deep and rigorous understanding
- A bet that models have underlying principles and structures that **generalise**

Personal Motivation (Neel Nanda)

Easy to get started, fast feedback loops, cross between maths, CS, natural sciences and truth-seeking. Code early and often — get contact with reality.

🍷 Case Study: Grokking ---



🍹 The Modular Addition Circuit ---



🍷 Frontier: Polysemanticity & Superposition ---

🍲 Polysemanticity: One Neuron, Many Concepts ---



🍷 Hypothesis: Polysemanticity is due to Superposition ---



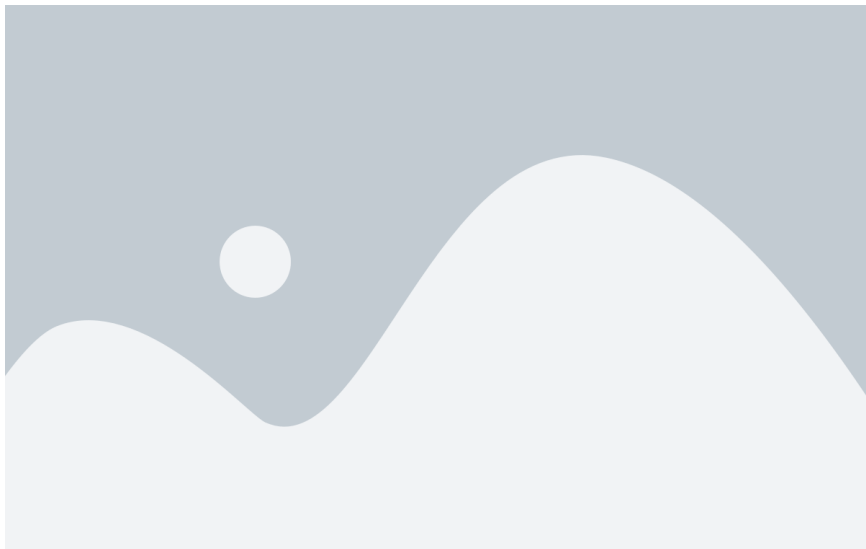
🍷 Geometry of Superposition ---



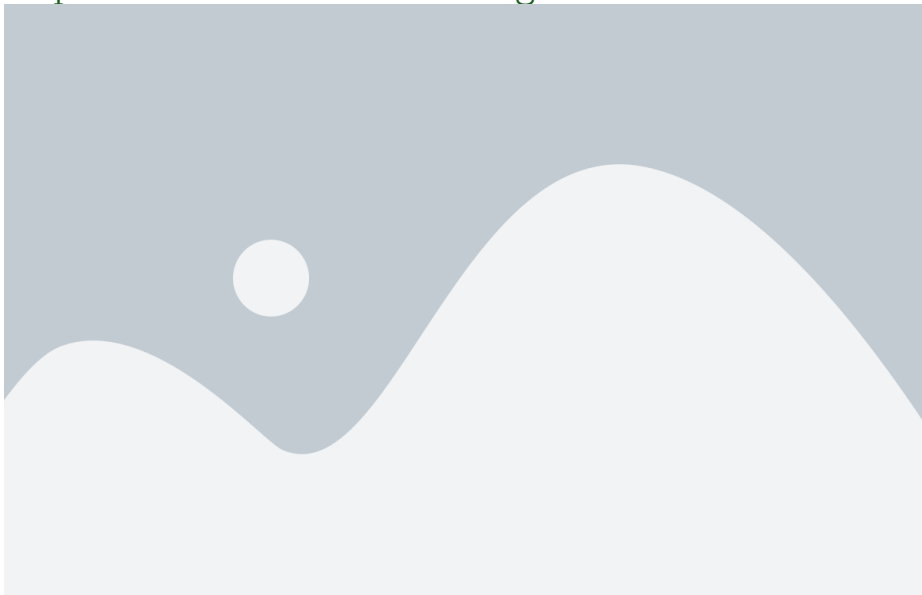
🍷 Case Study: Interpretability in the Wild ---



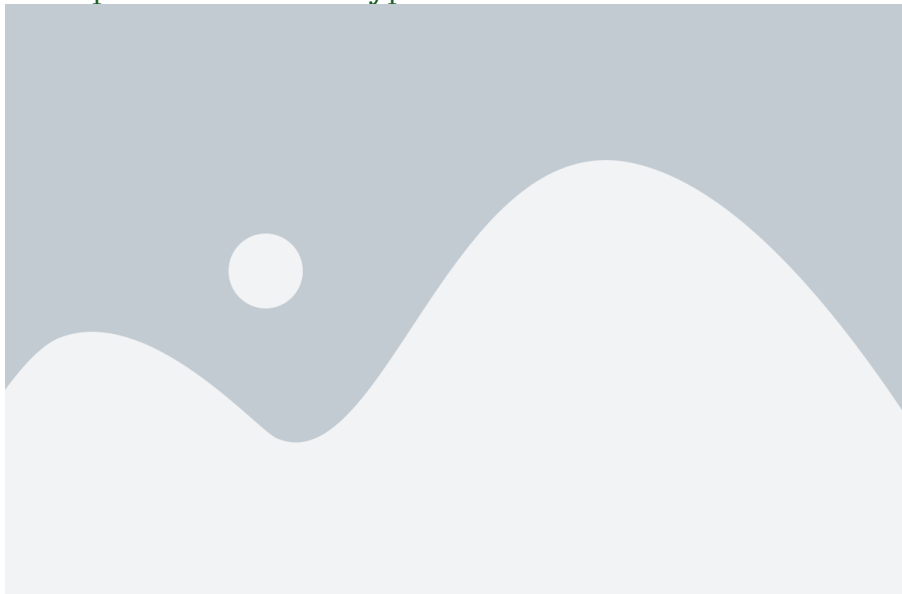
🍷 Refining Ablations: Mechanistic Interpretability as Validation ---



🍷 Technique: Activation Patching ---



🍷 Linear Representation Hypothesis ---



🍷 Case Study: Emergent World Representations in Othello-GPT ---



🍷 Othello-GPT: A Linear World Model ---



Learning More ---

- ****200 Concrete Open Problems in Mechanistic Interpretability****
neelnanda.io/concrete-open-problems - ****Getting Started in Mechanistic Interpretability****
neelnanda.io/getting-started
- ****A Comprehensive Mechanistic Interpretability Explainer****
neelnanda.io/glossary - ****TransformerLens**** (Python library)
github.com/neelnanda-io/TransformerLens

Open Problems covered in this talk

Studying Learned Features · Analysing Toy Language Models · Analysing Training Dynamics · Interpreting Algorithmic Models · Exploring Polysemanticity & Superposition · Finding Circuits in the Wild · Techniques, Tooling and Automation · Future Work on Othello-GPT

Thank You!

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- Elhage, Nelson, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, et al. 2022. “Toy Models of Superposition.” *Transformer Circuits Thread*.
- Elhage, Nelson, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, et al. 2021. “A Mathematical Framework for Transformer Circuits.” *Transformer Circuits Thread*.
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